



Factors Associated With Postpandemic Decline in US Motor Vehicle Mortality

Mathew V. Kiang, ScD; Monica J. Alexander, PhD

Introduction

Motor vehicle deaths surged during the COVID-19 pandemic,^{1,2} and remain a leading cause of death in the US. The national motor vehicle mortality rate peaked in 2021 at 13.0 deaths per 100 000 before declining 3 consecutive years to 11.5 deaths per 100 000 in 2024.³ Understanding factors of this decline is crucial to informing traffic-safety policies. We quantified how much of the decline reflects changes in driving distance, driving behavior, or fatalities among other road users.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

Methods

We decomposed the crude motor vehicle mortality rate change from 2021 to 2024 nationally and by state using 3 data sources. Death data came from the National Highway Traffic Safety Administration's Fatality Analysis Reporting System (FARS), a census of motor vehicle crashes on public roads where deaths occur within 30 days of the crash. FARS includes decedent demographics and restraint use, as well as crash-level behavioral indicators including excessive speeding or alcohol-impaired driving. State-level population estimates came from the US Census Bureau. State-level estimates of vehicle miles traveled (VMT) came from the Federal Highway Administration.⁴

We applied a Kitagawa decomposition⁵ to partition the mortality rate change into a VMT component and a per-mile risk component. We further partitioned the per-mile risk component into 11 mutually exclusive categories. Seven categories represent behavioral risk factors: speeding involvement, alcohol impairment, and unrestrained status (occupants without a seatbelt or helmet) in combination or alone. One category represents occupant fatalities not flagged by any of these behavioral risk factors. Three categories represent fatalities among other road users (pedestrians, bicyclists, other).

Uncertainty intervals (UIs) were obtained using Monte Carlo simulation where deaths were drawn from negative binomial, representing stochastic uncertainty, and VMT sampling error was incorporated using a lognormal model. All tests were 2-sided; statistical significance was defined as a 95% UI excluding 0 for national estimates and as FDR-adjusted $q < 0.05$ for state-level estimates (eMethods in Supplement 1). This study used public, deidentified data so was not considered human participants research by the Stanford University institutional review board, making informed consent unnecessary. Our study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for cross-sectional studies.

Results

We identified 166 230 motor vehicle deaths in the US between 2021 and 2024. Decedents had a median (IQR) age of 42 years (28–60 years); 45 953 (27.6%) were female and 119 851 (72.1%) were male. Over this period, the national crude mortality rate declined by 1.48 deaths per 100 000 (95% UI, –1.69 to –1.26). The VMT component was associated with 0.33 deaths per 100 000 (95% UI, 0.28 to 0.39) over this period, reflecting the increase in driving from 3.13 trillion vehicle miles in 2021 to

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3.29 trillion in 2024. This increase was offset by lower per-mile risk of mortality, which declined 1.81 deaths per 100 000 (95% UI, −2.03 to −1.59).

Unrestrained-only fatalities decreased by 0.42 deaths per 100 000 (95% UI, −0.50 to −0.34)—the single largest contribution and 23% of the per-mile risk decline (Table). Speed-only and alcohol-only fatalities each declined by approximately 0.10 deaths per 100 000 (speed only, −0.10 [95% UI, −0.16 to −0.03]; alcohol only, −0.11 [95% UI, −0.16 to −0.06]), while their combinations with unrestrained each declined by approximately 0.20 deaths per 100 000 (speed and unrestrained, −0.21 [95% UI, −0.26 to −0.15]; alcohol and unrestrained, −0.19 [95% UI, −0.24 to −0.14]). Fatalities not flagged by any of the behavioral risk factors declined 0.33 per 100 000 (95% UI, −0.43 to −0.23), or 18% of the per-mile risk decline. Pedestrian deaths declined 0.23 per 100 000 (95% UI, −0.31 to −0.14). Conversely, the bicyclist contribution (0.02; 95% UI, −0.01 to 0.05) and the other road user contribution (−0.05; 95% UI, −0.10 to <0.01) remained flat.

Motor vehicle mortality rate changes varied by state (Figure, A). Although VMT increased in 44 of 51 jurisdictions, no state experienced a statistically significant increase in mortality. The 3 largest

Table. Kitagawa Decomposition of the Change in US Motor Vehicle Mortality Rate From 2021 to 2024^a

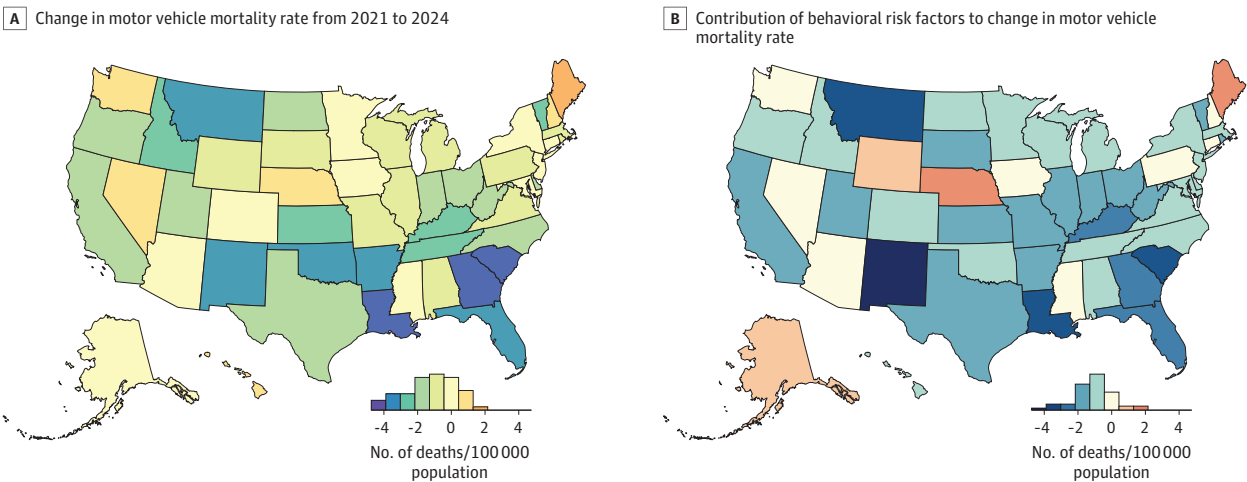
Component	Contribution to rate change (95% UI) ^b	Proportion of per-mile risk component, %
Total crude rate change	−1.48 (−1.69 to −1.26)	NA
VMT component	0.33 (0.28 to 0.39)	NA
Per-mile risk component	−1.81 (−2.03 to −1.59)	100.0
Per-mile risk components		
Behavioral risk factors	−1.23 (−1.38 to −1.07)	67.8
Speed only	−0.10 (−0.16 to −0.03)	5.3
Alcohol only	−0.11 (−0.16 to −0.06)	6.0
Unrestrained only	−0.42 (−0.50 to −0.34)	23.0
Speed and alcohol	−0.10 (−0.15 to −0.05)	5.5
Speed and unrestrained	−0.21 (−0.26 to −0.15)	11.4
Alcohol and unrestrained	−0.19 (−0.24 to −0.14)	10.5
All 3 flagged	−0.11 (−0.16 to −0.06)	6.1
No behavioral risk factor flagged	−0.33 (−0.43 to −0.23)	18.1
Other road users	−0.26 (−0.36 to −0.15)	14.1
Pedestrian	−0.23 (−0.31 to −0.14)	12.5
Bicyclist	0.02 (−0.01 to 0.05)	−1.2
Other (eg, truck, bus, ATV, etc)	−0.05 (−0.10 to <0.01)	2.8

Abbreviations: ATV, all-terrain vehicle; NA, not applicable; UI, uncertainty interval; VMT, vehicle miles traveled.

^a National Highway Traffic Safety Administration's Fatality Analysis Reporting System death counts and US Census Bureau population estimates were used to calculate crude mortality rates. VMT data came from the Federal Highway Administration.

^b 95% UIs were obtained using Monte Carlo simulation of 25 000 iterations and the percentile method.

Figure. Map of State-Level Decomposition of US Motor Vehicle Mortality Rate Change, 2021-2024



decreases occurred in Louisiana (−4.62; 95% UI, −6.75 to −2.53), Georgia (−4.22; 95% UI, −5.62 to −2.83), and South Carolina (−4.11; 95% UI, −6.34 to −1.89). Declines in specific behavioral risk factors had similar geographic variation (Figure, B); 13 states had at least 1 behavioral risk category with a statistically significant decline (Benjamini-Hochberg $q < 0.05$).

Discussion

The postpandemic decline in motor vehicle mortality reflects decreases in per-mile driving risk, not a reduction in driving. Fatalities involving an unrestrained occupant—alone or combined with speeding or alcohol—accounted for half (51%) of the per-mile risk decline, even as seat belt use remained stable near 91%,⁶ indicating the decline was not driven by increased belt use. Speeding-involved fatalities also declined even as self-reported speeding remained stable across 2019-2024,⁷ suggesting this decline was not explained by reduced speeding. Pedestrian deaths declined while bicyclist deaths did not, perpetuating a long-standing gap relative to peer countries.⁸ Sustained progress will likely depend on continued attention to occupant restraint and speeding, and on improving safety for cyclists and other road users,⁹ who saw no comparable decline.

Both VMT and FARS data quality vary by state. FARS 2024 data are preliminary and subject to revision. Uncertainty intervals reflect stochasticity in death counts and sampling error in VMT but do not capture systematic biases. Decomposition by sociodemographic factors such as age, race, and ethnicity was not possible because VMT is not stratified by these factors; therefore, we could not assess whether the decline narrowed the pandemic-era disparities concentrated among American Indian or Alaska Native and Black populations.¹⁰

ARTICLE INFORMATION

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Corresponding Author: Mathew V. Kiang, ScD, Department of Epidemiology and Population Health, Stanford University School of Medicine, 1701 Page Mill Rd, Palo Alto, CA 94304 (mkiang@stanford.edu).

Author Affiliations: Department of Epidemiology and Population Health, Stanford University School of Medicine, Palo Alto, California (Kiang); Departments of Sociology and Statistical Sciences, University of Toronto, Toronto, Ontario, Canada (Alexander).

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SUPPLEMENT 1.

eMethods.

SUPPLEMENT 2.

Data sharing statement